



Fig. 2: Optoelectronic sensing

Research has led to the development of various low-cost, low-capacity and more advanced electronic sensing devices such as electronic nose (e-nose).

An e-nose can detect and recognise odours and flavours with a high level of reliability. Using a number of electronic sensors, it can detect and differentiate between complex mixtures of volatile compounds.

The technology can be utilised to differentiate aromas of food and related materials.

Electronic nose technology, along with food adulteration detection technology, is successfully being used in the detection of plant diseases and pests. Some of its advantages are operational simplicity, non-destructivity and bulk sampling. However,

it comes with certain drawbacks as well, such as low sensitivity and specificity in comparison with microbiological and molecular methods.

e-Nose technology is still in development phase, both with respect to hardware and software. e-Noses contain an array of sensors (from one to 32), using a variety of sensor technologies like organic polymers and metal-oxides to micro-

balances. These are being widely used by some companies as quality-control instruments. Their strengths include high sensitivity and correlation to human sensory panels for many applications.

Limitations to their full potential includes loss of sensitivity in the presence of water vapour or high concentrations of a single component like alcohol, sensor drift and the inability to provide absolute calibration, relatively short life of some sensors, necessity to do considerable method development work for each specific application and lack of being able to obtain quantitative data for aromatic differences. There is some evidence that sensors differentiate aromas based on relatively few compounds. In the future, a relationship between specific chemicals and a single flavour attribute may be achievable.

Optoelectronic sensing

Optoelectronics is the discipline of electronics that focusses on light-

TAIYO YUDEN Wireless Module Product Lineup

Personal Area Network

Bluetooth® LE (Basic Type)

BT4.2 - Cortex-M0 (nRF51)

EYSGCNZWY EYSGJNZWY EYAGJNZXX
EYSGCNZXX EYSGJNZXX



BT4.2, STD, Antenna, 32KHz clock DC/DC
BT4.2, Small, Antenna
BT4.2+ANT™, Small, Antenna

BT5.0 - Cortex-M4/M4F (nRF52)

EYSHCNZWZ EYSHJNZWZ EYSHSNZWZ



BT5.0+ANT™, STD, Antenna, 32KHz clock DC/DC
BT5.0+ANT™, Small, Antenna
BT5.0+ANT™, Ultra Small, Antenna

EYSKBNZWZ EYSLCNZWZ



Full BT5.0+802.15.4 (Thread) High performance
BT5.0, STD, Antenna, Simple

Under Development

Local Area Network

WLAN: IEEE802.11 SDIO

2.4GHz (11b/g/n)

WYAGVXD7 11b/g/n, Low-P, Antenna

2.4GHz+5GHz (11a/n/ac)

WYSBHVXG 1x1 11ac, BT4.2, SMD

WYSAGVXD7 WYSEGVXD7



1x1 11ac, BT4.2, SMD w/ Antenna
Under Development
1x1 11ac, BT4.2, SMD w/ RF connector

WLAN: IEEE802.11 SW Embedded

WYSACVLXY-XX 11b/g/n + CPU, SMD, Antenna, Cortex-M4, TY app.



Navigation System

GPS/GNSS

GYSFFMAXB



GPS with CPU, Memory, SMD

GYSFFMAXC



GNSS with CPU, Memory, SMD

GYSFDMAXB



GPS with CPU, Memory, Antenna,

- Production
- New, U/D items

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